Mapping the Shape of the U.S. Economy: A Topological Data Analysis Approach with Ball Mapper

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ABSTRACT

Topological Data Analysis (TDA) is a developing data analysis method. Currently, a large body of TDA research utilizes the traditional Mapper algorithm. This paper aims to show an entry-level path into TDA, and to expand the body of literature using Ball Mapper–an algorithm adjacent to traditional Mapper. Using U.S. macroeconomic data from the Bureau of Economic Analysis (BEA), the Federal Reserve Bank (FRB), and the Bureau of Labor Statistics (BLS), we begin by showcasing Ball Mapper's use in exploratory data analysis. We then show how we can effectively use Ball Mapper in combination with other derived economic measures: the Real Interest Rate, PPI-CPI Gap, and Real Wage Growth. Our results reveal an underlying structure in the U.S. economy from 1960 to 2024. Illustrated through Ball Mapper's graphs, we find that economic expansion contains distinct stages, that peaks and troughs are structurally separate from the broader expansion cycle, and that no two economic contractions are alike. Notably, despite not being included as an input, our analyses uncovered meaningful relationships between years, which suggests that the relationships seen are inherent to the economic data itself.

KEYWORDS

Topological Data Analysis; Topology; TDA; Ball Mapper; Mapper; Data Science; Economics; Macroeconomics

INTRODUCTION

Topological Data Analysis (TDA) is an emerging field of data analysis that offers a unique vantage point for exploring complex data. Broadly, traditional TDA applications use two tools: an algorithm called Mapper, whose output is represented by a mathematical graph, and an analysis technique called Persistent Homology.[1A] These two tools are not necessarily dependent on one another and both provide information about data in their own right. Moreover, Mapper and Persistent Homology are central to TDA's core argument: data contains an underlying shape, and this shape can provide us with qualitative, and sometimes quantitative, insights about large multidimensional datasets.[2]

At a high level, Mapper is used to explore and visualize high-dimensional data. Briefly, the Mapper algorithm achieves this by producing a mathematical graph using a filter function. This graph presents a picture analogous to taking an x-ray of the data, revealing the skeleton of the dataset; it allows one to find structural relationships that are not obvious at first glance. Persistent Homology's goal is to uncover topological features of a dataset (components, holes, voids, etc.) through various scales. As an example, think about viewing a small picture made of dots as you move it from touching your nose to arm's length away. Too close, you only see individual dots; too far away, the dots merge together into a singular shape. Persistent Homology takes observational notes about how the picture changes across all the various distances.

Moving forward, we will use the terms "traditional Mapper" and "Mapper" interchangeably. When we use the term "traditional Mapper," we are referring to the methodology for Mapper presented by Singh et al.[3] in 2007. This is an important distinction because the concept of the Mapper methodology has been implemented in various ways by changing the filter function and other parameters.

Although Persistent Homology provides insight into the topological features our data contains, the focus of this paper is on the structural exploration that Mapper and Mapper variations offer. Specifically, we are examining various macroeconomic indicators of the

[A] A mathematical graph, from Graph Theory, is a diagram that consists of nodes and edges.

United States (U.S.) using a Mapper-adjacent algorithm called Ball Mapper (BM). One advantage of BM is that it requires fewer parameters than traditional Mapper, allowing for easier implementation.

In traditional Mapper, one must pass the data through three stages, all with their own parameters, in order to generate a graph. Ball Mapper, however, only needs one parameter before producing a graph.[4] An important trade-off between Ball Mapper and traditional Mapper is that while BM reduces the number of steps, one loses the ability to fine-tune outputs. Additionally, traditional Mapper literature presents itself as requiring a mathematical background, such as an introductory course in Topology, or conceptual knowledge of data science methods and algorithms. Although it is not impossible to learn TDA without the above backgrounds, TDA might seem unnecessarily complicated for data analysis, or impossible to decipher.

One major motivation for this paper came from a paper by Dłotko et al.[5] The paper was a two-pronged macroeconomic analysis using Ball Mapper. The first prong used a dataset of five macroeconomic variables from 16 countries from the 1870s to 2017. The authors were interested in comparing how countries have evolved over time, their transformation from the Great Depression Era, and various views on wealth and inequality. The second prong was to look at the relationship between private credit growth and the GDP of various countries. This paper was the only one in the literature that we could find that combines both macroeconomics and Ball Mapper.

This analysis will expand the small body of literature whose main focus is applications with Ball Mapper.[1] We also hope that it will serve as an on-ramp for anyone interested in TDA who feels inundated with jargon during early-stage research.

Using U.S. macroeconomic data from the Bureau of Economic Analysis (BEA), the Federal Reserve Bank (FRB), and the Bureau of Labor Statistics (BLS), we plan to achieve these objectives in two stages. The first stage is to show Ball Mapper's use in exploratory data analysis (EDA) by looking at structural observations, coloration, and the size variation of nodes in our graph. Then, using what we demonstrated in our first stage, we extend that framework to other derived macroeconomic measures: the Real Interest Rate,[B][6, 7] PPI-CPI Gap,[C][8] and Real Wage Growth.[9][D]

METHODS AND PROCEDURES

Data Selection & Preparation

We pulled data from three publicly available U.S. government sources: The Bureau of Economic Analysis (BEA), The Federal Reserve Bank (FRB), and The Bureau of Labor Statistics (BLS).[10, 11] We gathered the data using R via two application programming interfaces (APIs): one for data from the BEA, and the other from the Federal Reserve Economic Data (FRED) API.[12, 13][E] We additionally used recession dates based on business cycle contractions and expansions provided by the National Bureau of Economic Research (NBER). These agencies were selected because they are authoritative sources for U.S. economic data. Their widespread use in both the private and public sectors gives us high confidence in the accuracy and integrity of the data.[14]

Data Series	Series Abbreviation	Year Start	Year End	Frequency	Series	API
Gross Domestic Product	GDP	1930	2024	Annual	Table 1.1.1	BEA
Personal Income & Its Disposition	PID	1948	2024	Annual	Table 2.1	BEA
Foreign Direct Investment	FDI	2014	2024	Annual	Direct Investment and MNE	BEA
Federal Funds Rate	FFR	1955	2025	Annual	RIFSPFFNA	FRED
Employment Cost Index	ECI	2001	2025	Quarterly	ECIALLCIV	FRED
Consumer Price Index	CPI	1947	2025	Monthly	CPIAUCSL	FRED
New Privately-Owned Housing Units Started	Housing Starts	1959	2025	Monthly	HOUST	FRED
Producer Price Index - Finished Goods	PPI	1947	2025	Monthly	WPSFD49207	FRED
Unemployment Rate	Unrate	1948	2025	Monthly	UNRATE	FRED

Table 1: Summary of data series, time period, and sources for this analysis.

Table 1 shows all the data series we considered for this analysis. We excluded Employment Cost Index (ECI) and Foreign Direct Investment (FDI) due to their limited availability of years. Of the remaining data series, New Privately-Owned Housing Units Started

[B] Approximation using the Fisher Equation.
[C] Producer Price Index: Finished Goods less Consumer Price Index.
[D] Compensation less Inflation (CPI).
[E] FRED aggregates data from national and international sources, as well as public and private sources.

(Housing Starts) had the smallest range and so set our lower bound for the range years used (1960-2024). Our upper bound was the latest full calendar year that was available. We wanted to represent the economy from various vantage points, i.e., the broad economy, policy makers, consumers, business owners, etc. Table 2 shows the data series that we felt were representative of the economy as a whole and had data for our time period of interest. Within each of these series we used a subset of columns to construct our final dataset for analysis. To see what parts of the economy we determined each series plays, the column Functional Description summarizes them and can be found in Table 2.

For this analysis we are focusing on an annual time frame and data transformation was needed. Looking at Table 1, not all our data was provided on an annual basis. Table 2 shows a summary of the transformation taken and the details for each follow.

The first transformation we did was on Personal Income and Its Disposition (subsequently, Personal Income). This data is reported in nominal dollars and does not account for inflation. So, to account for inflation we used the Consumer Price Index (CPI) to transform our values from nominal dollars to real dollars (Equation 1).[15] We then took the log difference (Equation 2) from the output from Equation 1 to get an approximate percentage change from the preceding year.

$$\text{Real Dollars} = \frac{\text{Nominal Dollars}}{\text{CPI}} \times 100 \tag{1}$$

Calculating the year-over-year percent change using log differences provides us with a few advantages. First, taking the log stabilizes the variance and makes exponential trends across the series more linear. Second, the difference between log values approximates the true percentage change well for small to moderate values. Further, the log difference allows for symmetric percent changes: increases or decreases from year to year are of equal magnitude.[16] An interesting question that has not been explored to our knowledge is the impact on linearized data and Ball Mapper's distance metric, the Euclidean distance. Further exploration of this question will be touched on in the *Discussion* section.

$$\Delta \ln(\text{Level}) = [\ln(\text{Level}_t) - \ln(\text{Level}_{t-1})] \times 100 \tag{2}$$

Our next transformations were applied to housing starts, Producer Price Index - Finished Goods (PPI), and CPI. All three of these sources were only provided on a monthly time frame, so to get an annual value for them we used the simple arithmetic mean. Once these data were in annual form, we then found the percent change from the previous year using the log difference (Equation 2).

Housing starts data is given in a seasonally adjusted annual rate (SAAR). To get this number the U.S. Census first takes the raw monthly number of housing starts reported to their survey. Then they adjust that number to account for seasonal patterns (e.g. summertime booms or winter lulls), and finally they multiply it by 12. How we interpret this final number is: for the reporting month, it is the number of houses that would be built at the end of a 12-month period.[17]

Series Abbreviation	Economic Role	Functional Description	Transformation Applied
GDP	Growth Composition	Consumption, Investment, Gov., and Trade	None (Source in % Change)
PID	Income Structure	Wages, Entrepreneurship, Gov. Transfers	Real Adjustment & Log Difference
UnRate	Labor Market Dynamics	Labor Distress Level and Trend	None (Annual Rate) & Simple Difference
FFR	Monetary Policy	Cost of Capital and Borrowing Conditions	None (Annual Rate) & Simple Difference
PPI	Supply-Side Signal	Producer Costs & Measures Cost-Push Inflation	12-month Avg. & Log Difference
CPI	Demand-Side Signal	Cost of Living & Measures Demand-Pull Inflation	12-month Avg. & Log Difference
Housing Starts	Leading Indicator	Physical Residential Production	12-month Avg. & Log Difference

Table 2: All years for analysis are 1960-2024.

Both PPI and CPI are economic indexes that represent a weighted average of goods they respectively track. Although PPI has many different series, the one used for this analysis tracks the price the producer receives from the physical goods that it sells to businesses or consumers. An important distinction to note about the PPI used here is that the index captures goods which are finished or final in the economic sense. Colloquially, it is similar to the wholesale price a retailer might pay to a producer before marking up for retail sales. However, it also includes goods which are sold to businesses, think fully assembled computers or fleet vehicles.[18] CPI on the other hand captures the consumer side and the retail price. That is, the cost of the good after profit margin, transportation costs, and other intermediary costs are calculated into a goods price sold at the register.[19] Broadly, both have been associated with measuring inflation for

producers and consumers. The more nuanced understanding is that index changes can reflect many things: cost-push inflation, demand-pull inflation, supply shocks, monetary policy inflation, and wage-price spirals are just a few examples. It is our hope that Ball Mapper might show the distinctive separation in the various aforementioned reasons for the PPI and CPI indexes changing from year to year.

Finally, after all the transformations are complete, we used a total of 16 variables from our selected data series (Table 2) for our analysis. Again, we aimed to use a comprehensive set of variables to represent the U.S. economy. A list of variables used can be found in Table 3 in no particular order.

PPI Finished Change	Personal Income	Compensation	Entrepreneurship
Transfers	Consumption	Domestic Investment	Government Spending
Exports	Imports	Inflation	Unemployment
Unemployment Change	Fed Rate	Fed Rate Change	Housing Change

Table 3: Variables inculde in Ball Mapper Point Cloud

Ball Mapper

In-depth descriptions of Ball Mapper's theory have been covered in various papers.[4, 20, 21] As the aim of our paper is to provide an introductory foray into Ball Mapper and TDA in general, we will omit some of the more technical details below.

Theoretically, Ball Mapper supposes we are given a dataset X with K dimensions, N observations and some metric d. Then to create a Ball Mapper graph, our goal is to construct a collection of balls, let's call it B, with a constant radius, ϵ, such that every observation of our dataset is covered by at least one ball in B. Mathematically, this is written as follows: Let (X, d) be a metric space where $X \subset \mathbb{R}^K$ is our dataset and d is a distance metric on X. For a fixed radius $\epsilon > 0$, let $C \subseteq X$ be a set of center points. Then for each $c \in C$, we define a ball $b(c, \epsilon) = \{x \in X : d(c, x) \leq \epsilon\}$. Our collection then is $B = \{b(c, \epsilon) : c \in C\}$ and $\bigcup_{b \in B} b = X$.

In practice, our dataset X is called a Point Cloud. Our N observations or rows are normally referenced as points, and our K dimensions, the columns of X, are denoted variables. Since Ball Mapper is known to handle large multidimensional datasets well, it is common to have point clouds with dimensions $K > 2$. Ball Mapper was originally written in R, but later on was adapted for use in Python. For this analysis we used R and the Ball Mapper package aptly named "BallMapper".[22]

To construct the Ball Mapper (BM) graph, the balls of our graph are created in an iterative manner and not a predetermined manner like the theory defines. If we pre-defined a subset of ball centers, it would take away from the driving idea behind TDA–that data has shape–as well as start moving into the territory of other clustering methods such as K-Means Clustering, where you select a number of clusters based on statistical properties of the data. This iterative process allows our point cloud to cluster in a "natural" way based on our choice of epsilon and distance metric.

$$d(x_i, x_j) = \sqrt{(x_{i_1} - x_{j_1})^2 + (x_{i_2} - x_{j_2})^2 + ... + (x_{i_k} - x_{j_k})^2} \leq \epsilon \tag{3}$$

The iterative ball creation is carried out in a linear fashion from the first row to the last of the point cloud. The result of this linear selection process is a greedy, sequential creation of ball centers. How this works is that it takes the first point, let's call it c_1, of the point cloud and assigns it as the center of the first node in our graph. Then using the default distance metric, the Euclidean Distance (Equation 3), it compares every point in the point cloud to c_1 to create $b_1 = b(c_1, \epsilon) = \{x \in X : d(c_1, x) \leq \epsilon\}$. After this first ball is created, Ball Mapper looks for the next row in the point cloud that is not a member of the first ball, and assigns it as the next center. This process continues until all the points are in a ball and the union of balls covers our whole point cloud.

Hopefully, it is clear now why this process can be described as greedy and sequential. It relies a great deal on the point cloud order as well as the epsilon. In fact, the Ball Mapper graph changes for epsilons sufficiently large enough to change at least one point's inclusion or exclusion to a ball; the graph can also change for unique orders of the point cloud rows. We touch on this in greater detail during our *Discussion*.

The last step to the Ball Mapper graph is to draw the edges if they exist. The way the BM algorithm is structured, it allows for the possibility of points to be in multiple balls. When two balls (also called nodes) contain the same point, they are said to intersect. If two nodes share at least one point, then an edge is created. If there is more than one shared point, the number of shared points dictates the

edge strength. So, once we have all the nodes, edges, and edge strengths, we then can construct a Ball Mapper graph. The Ball Mapper algorithm described above can be broken down as follows:

1. Select a point $x_i \in X$ and assign it as c_i.
2. Given $\epsilon > 0$, construct a ball $b_i(c_i, \epsilon)$ with a center c_i and radius ϵ by checking if x_j satisfies $d(c_i, x_j) \leq \epsilon : x_j \in X$ and d is the distance metric (Equation 3).
3. Place b_i in a set B.
4. Select another x_n and repeat steps 1–3 such that $x_n \neq x_j : x_j \in b_i$.
5. Draw an edge between $b_i, b_j \in B : b_i \neq b_j$ if $b_i \cap b_j \neq \emptyset$, weighting the edge based on the size of the intersection.

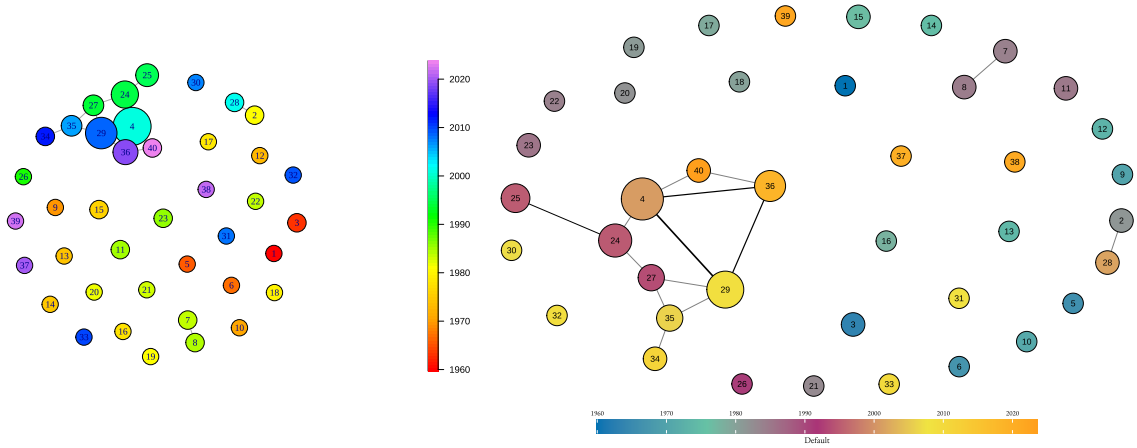
Implementation

To use Ball Mapper one only needs three parts: a point cloud, a variable to color by, and an epsilon. For this analysis, our point cloud was made up of 16 variables and 65 points, each representing a year from 1960–2024. Two of the variables, *Unemployment Change* and *Fed Rate Change*, were created before any transformation took place. We elected to create these because the Unemployment Rate (Unemployment) and the Federal Funds Rate (Fed Rate) were the only two dimensions which did not have a transformation and thus did not have a year-over-year change like our other variables. Having the rate of change for both Unemployment and the Fed Rate is also important because it gives us additional information beyond the intrinsic value of each of the measures. Both measures on their own give us a current look at the economy: the Unemployment Rate telling us how many who are not working but want to,[23] and the Fed Rate signaling the structural cost of capital.[24]

There seems to be no generally accepted way of choosing epsilon in the literature. Since Ball Mapper constructs maps by selecting a point and creating a ball with a radius of epsilon, it is possible to select an epsilon small enough to produce a graph where every point is a node of only itself. Conversely, this should lead to an epsilon big enough such that all of our points are included in one node. For our analysis here, this was indeed the case. So to find an appropriate epsilon we first started by finding lower and upper bounds.

What we look for in our lower bound is a graph with an epsilon small enough to make every point in our point cloud a node of one point, a singleton node, not connected to any point. For our upper bound we similarly look for an epsilon so large that it creates a singular node that houses all of our points. Additionally, to reduce some of the computational time one can look for a lower bound where a connected component starts to form, and an upper bound where we have a connected component of only two nodes. This cuts off tail end graphs that generally don't provide much substantial insight into the data.

For our analysis we found an initial lower and upper bound represented by the interval $[0.38, 0.90]$. We then wrote a function to generate around 100 graphs, reviewed them, and narrowed the upper and lower bounds to $[0.40, 0.70]$. During this narrowing process we were looking for interesting features: connected components forming or dissolving, flares coming off of any components, and notable sizing or coloration patterns. The resulting value of this process was 0.474. We thought it contained all the aforementioned features. This final epsilon is used for all the graphs generated in the analysis.



(a) Ball Mapper graph using standard function, ColorIgraphPlot.

(b) Ball Mapper graph using new graph output created with ggraph.

Figure 1: Ball Mapper graph colored by Year; epsilon set to 0.511.

Before describing some visual changes we made to the Ball Mapper output graph, it is important to note that Ball Mapper will output the same graph unless at least one of two things changes: the size of the point cloud changes or the order of the point cloud. The former intuitively makes sense. The algorithm is creating epsilon radius balls so adding or subtracting any data will yield a different amount of points in at least one ball. Depending on the data being changed or number of points, the structure of the output might not change much. As an example, when we were deciding on which PPI series to use, we had a multitude of choices. Initially we used PPI - All Commodities but when reading the BLS handbook on PPI, it specifically mentions that PPI - All Commodities is susceptible to double counting. To avoid this issue we used PPI - Finished Goods instead.[18]

The standard Ball Mapper package in R produces a sufficient graph for analysis. The output object of the Ball Mapper algorithm is a series of nodes, edges, edge weights, and other graph properties created by the package *igraph*.[25] Separate from the igraph object, a graphing function that uses R's base plot function and default coloring scheme produces the visual graph. This is an advantage because when packages are written in Base R they are usable to anyone who can install R. We, however, felt that this graph output could be improved upon and recognized that we could use alternative, and more comprehensive packages to create a graph, *ggraph*.[26]

Using the same underlying output igraph object from Ball Mapper, we created the same graph in ggraph but changed the visual representation and added another helpful piece of information, edge weight, that was given but not used in the default BM graph. Looking at Figure 1, we have the standard Ball Mapper graph output on the left (Figure 1a), and our new ggraph output on the right (Figure 1b). The most prominent changes are the spread of the network and coloration.

Looking at Figure 1b, having the nodes spread out helps us see the structure of our graph more clearly as well as the edge weight between nodes. We see that highlighting the edge strength presents a potential insight between certain nodes. Although our data is relatively small, consisting of 65 rows, one could imagine the advantage Figure 1b might have with a data set much larger than this one (e.g. looking at our time frame on a quarterly or monthly basis). Additionally, we changed the color palette for accessibility and readability.

For Figure 1b and Figure 2, we use Year as our initial coloring variable. We started with this variable because Dłotko et al.[5] provided a framework for this type of analysis. In this paper the authors started by coloring their graphs by Year to initially understand the shape of their data. Later, in the second half of their analysis, they produced more graphs each colored by the point cloud's variables. Since we have 16 dimensions to our point cloud, an analysis of that size is outside the scope of this paper. An additional note about Dłotko et al., when they colored by the variable 'Year', it was not a variable in their point cloud. Mentioned in the *Introduction*, we recognized the usefulness of this technique and extended this idea to our point cloud. Specifically, we calculated variables from our data that we thought represented different economic lenses: the Real Interest Rate (Figure 3), PPI-CPI Gap (Figure 4), and Real Wage Growth (Figure 5).

The Fisher Equation (Equation 4) is a mathematical relationship to approximate the Real Interest Rate. It captures the lived experience of purchasing power for producers and consumers by showing what is left of a nominal interest rate after accounting for inflation–in our case we are using the nominal Federal Funds Rate (Fed Rate). When the real interest rate is negative, inflation is higher than the Fed Rate and is sometimes described as the Federal Reserve (the Fed) being, "behind the curve."[27] Being behind the curve can stem from various factors but some common interpretations are, the Federal Reserve does not want to try and tame inflation because of the possible negative impact on the labor market, or the Federal Reserve is hopeful that the economy will grow into the inflation, thereby raising wages up to or beyond inflation. On the other hand, when the real interest rate is positive, the Fed Rate is higher than inflation and could signal that the Federal Reserve is cautious about inflation's impact on the economy. This is described as being "ahead of the curve." If the Fed Rate is ahead of the curve, it indicates the cost of capital is higher causing businesses to "feel" the rate. In reaction to business being more expensive to conduct, employers might cut costs or freeze hiring, directly affecting the labor market as a result of keeping prices stable. An important note about the Fisher equation: the approximation does not hold well when nominal interest rates (Fed Rate) are relatively high and when time spans are short.[7]

$$\text{Real Interest Rate} \approx \text{Fed Rate} - \text{Inflation} \tag{4}$$

We calculated the PPI-CPI Gap by taking the difference between PPI - Finished Goods and CPI (inflation). As a quick reminder, PPI represents an "at-cost" revenue (before adding margin, transportation, *etc.*) that producers receive, and CPI is a common measure of consumer inflation. It is important to note that although CPI is used for general consumer inflation, the Federal Reserve prefers Personal Consumption Expenditures (PCE) because it tracks more normal consumer behavior such as substitution of products. The predominant question for this measure was: if producers or consumers are experiencing inflation, can our graph show the primary driving force behind the complex nature that is inflation?

Similar in nature to the PPI-CPI gap, we took the difference between Compensation and Inflation as an approximate measure for Real Wage Growth. With this measure, we were interested in the relationship between people's annual compensation and inflation, i.e., are

wages keeping up with inflation. Generally, for this relationship, positive numbers indicate that wages are growing faster than inflation, and vice versa if it is negative–effectively people are taking a pay cut.

RESULTS

Exploratory Data Analysis

Figure 2 shows three different connected components and many nodes not connected to any others. A connected component is when at minimum, two nodes share one edge. If a node is not connected to any other node, they are called singletons. When a node is a singleton, it suggests that node is unlike all the other data points and indicates that it is a possible outlier in the dataset.[5] Since our graph consists of 40 nodes and three components, we will provide all the singletons in a table (Table 4). For the three connected components, we see there is one large and two small components. Moving clockwise, starting with the large component on the left-hand side, we will label them C1,[F] C2,[G] and C3[H] for ease of reference. To prime the reader, the following section briefly discusses interpretation of Ball Mapper graphs via visual inspection. We then apply our knowledge to our three derived metrics to finish the section. Afterwards, we move to the *Discussion* section where we investigate the contents of our nodes and R is a necessary tool.

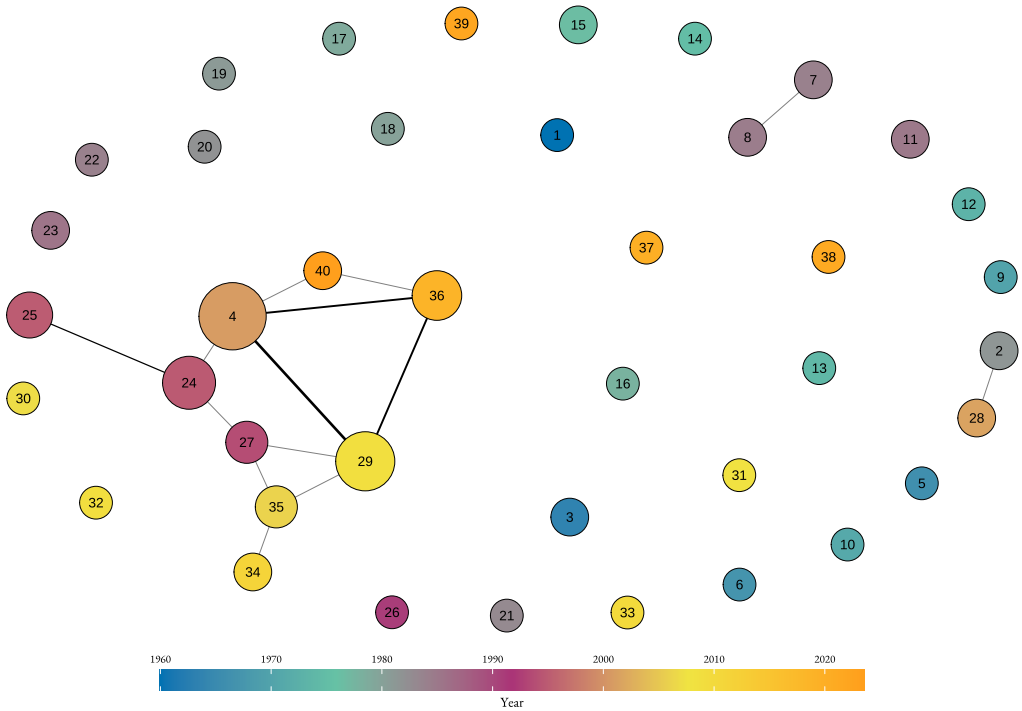


Figure 2: Ball Mapper graph colored by Year; epsilon set to 0.474.

When doing a visual inspection of Ball Mapper graphs, there are essentially two categories to view it through: coloration and shape. Most often, the coloration of the Ball Mapper graph is used to highlight a variable of interest, or certain known structures in the data. This could take the form of highlighting known outliers, showing a coloration pattern (e.g. a smooth gradient from one end of the graph to the other), or a combination of the two; i.e., nodes which do not conform to patterning. Figure 2 shows us no global coloration pattern, but does reveal four sub-groups of colors in C1 as well as the mismatching coloration for C3. From visual inspection, C1's four sub-groups of colors show distinct time periods: 1990s (magenta), mid-2000s (yellow), 2020s (orange), and 2000 (light terracotta). We also notice that for C3, 2000 and 1980 are what time periods its nodes show, while C2 shows a consistent mid-1980s coloration. Looking at shape and structure of the graph, as mentioned above, nodes contain an edge between them when there is an intersection of at least

[F] Nodes: 4, 24, 25, 27, 29, 34, 35, 36, 40
[G] Nodes: 7 and 8
[H] Nodes: 2 and 28

one point. Additionally, the edge weight is commensurate with how many points are in the intersection, and the overall size of the node indicates the number of points it contains. Particular to C1, nodes which are not directly connected by an edge (e.g. Nodes 24 and 36), can be taken as similar in some fashion, but also signal a significant difference within the component.[5]

Examining C2 and C3 closer, C2's nodes are colored very similarly, and are of the same size. As a relative measure, using Table 4, we look at our singletons and notice that many only hold one point. Thus, comparing C2 and C3 nodes, we should expect around one to two points in each of the nodes. C2's nodes contain the years 1968 and 1969 in nodes 7 and 8, respectively, and 2000 is in both, creating the edge. Mentioned previously, C3's coloration is not consistent. Node 2 contains 1961 and node 28 contains 2001 with 2002 being our bridge year. What is an interesting observation that will come up again when looking at C1 is how different, but grouped together the years are. This brings up questions as to why 1961 and 2001 would be seen as similar and is explored in the *Discussion* section.

Structure-wise, C1 is the largest component and contains nine total nodes. We mentioned already that it contains three subgroups of colors and observed the indicated years they visually represent. Of the nine nodes that make up C1, we are provided two distinct parts: a large triangle made of the largest nodes with strongly connected nodes (4, 29, and 36); and a sweeping arm to the left of the triangle that contains leaves on each end (nodes 25 and 34).[I] The combination of structure along with the subsets of the colors immediately present a notable result. On the upper half of our arm we see nodes 25, 24, and 27 all colored similarly. We have a similar outcome for the bottom half of the arm and one node from our triangle. Conversely, and equally interesting, the triangle's nodes are three distinct colors, while also being the largest nodes in the graph and containing the strongest edge strengths between them.

Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)	Node	Year(s)
1	1960	**10**	1971	**15**	1976, 1977	**20**	1982	**30**	2007	**38**	2021
3	1962, 1965	**11**	1972, 1998	**16**	1978	**21**	1983	**31**	2008	**39**	2022
5	1966	**12**	1973	**17**	1979	**22**	1984	**32**	2009		
6	1967	**13**	1974	**18**	1980	**23**	1985, 1986	**33**	2010		
9	1970	**14**	1975	**19**	1981	**26**	1991	**37**	2020		

Table 4: Singleton nodes and the years they contain.

Although we would need to investigate what points make up the nodes to gain more insight into our data, on the surface what we can conclude about C1 is: the sweeping arm seems to have two groups representing two time periods in our range of years. We also notice that nodes 27, 35, and 34 make their own small cycle of nodes but are a mix of the two aforementioned time periods. Lastly, our triangle has a combination of factors which warrant further investigation: the nodes are the largest in size indicating they contain the most points; each pair of the nodes (4, 29, 36) contain the strongest edge strengths of the whole graph; all three nodes are distinct colors indicating separate time periods; and finally, the distinct colors seem to contain a path through time going from 2000 (node 4) around in a counter-clockwise direction to present day (2020–2024).

Real Interest Rate

Figure 3 shows us that colors representing blue/mint green are negative values which represent inflation far outpacing the Fed Rate. In contrast, values colored yellow/orange show us that Fed Rates are very high compared to inflation. In general we notice that there are not many nodes which are blue or mint colored, and most nodes range from purple to yellow (approximately -2 to 4). Looking at the extreme ends (less than -4 or greater than 6), we see there are about four nodes for each end. On the low end are nodes 39, 14, 38, and 34; and on the high end are nodes 19–21. The reader may have noticed both groups of nodes seem to be consecutive (or close to), this is not a random event but a feature in Ball Mapper. We touched on this in the Methods and Procedures: Ball Mapper section, and will revisit this idea in our *Discussion*. Lastly, and the only case of a coloration gradient, C1 shows a smooth transition from negative values at the bottom (34) to positive in a sweeping fashion like our arm.

PPI-CPI Gap

Figure 4 shows a very different distribution of colors compared to Figure 3. Overall, a majority of our nodes are below the midpoint of the legend. This shows that regardless of the driver behind inflation on both consumers and producers, consumer inflation tends to outpace producers. Additionally, our legend is not balanced like it is in Figure 3; this supports our clustering of values below the middle of our legend. In fact, we see that many nodes seem to be from -2 to -1. With a discerning eye, we estimate to have three nodes which are at the very end of our negative values (nodes 23, 37, and 9). However, on the opposite end there is only one node that appears to be an extreme positive, node 39.

[I] The term leaf (leaves) could also be called a flare. How we define flare is two or more nodes in a row that are hanging off the main body of a component.

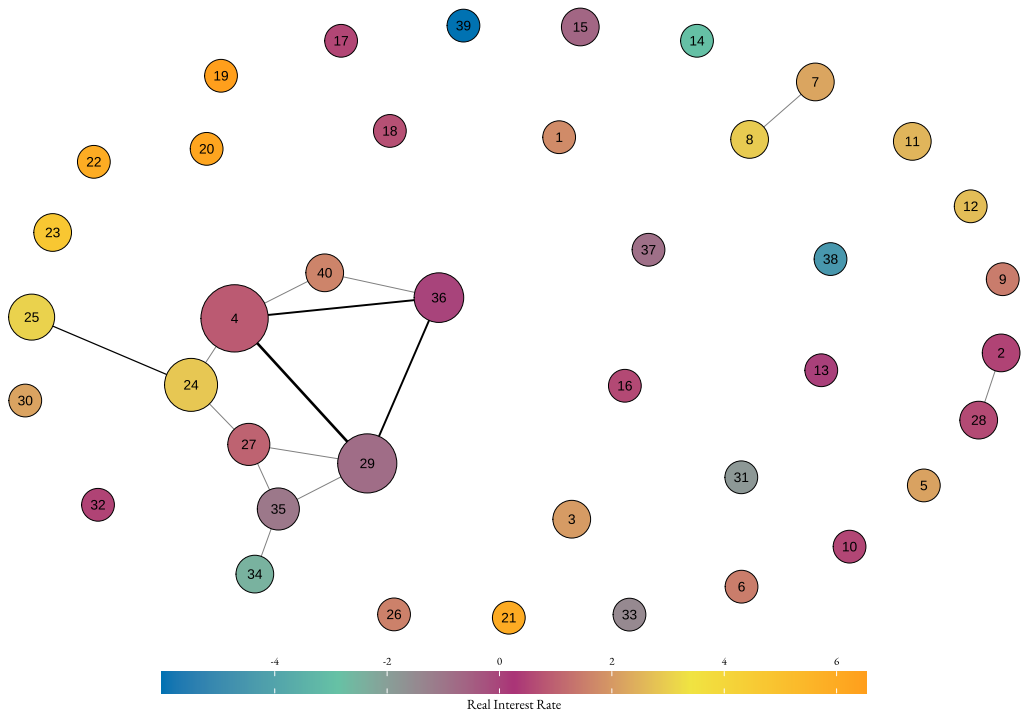


Figure 3: Ball Mapper graph colored by the Real Interest Rate; epsilon set to 0.474.

Real Wage Growth

Similar to Figure 4, there is a clear concentration of values on one end of our legend Real Wage Growth shown in Figure 5. Using the center of the legend as a reference point, we can see that Figure 5 is even more skewed than Figure 4; its zero starts two tick marks from the center instead of one. To give a range for the overwhelming yellow coloring, most values seem to fall between -2.5 and 2.5, with most of these values lying right around zero. With many values indicating zero, our selected time period tells us that wages have seemed to keep pace with inflation. Although Figure 5 and Figure 4 are showing different measures, Real Wage Growth shows a much longer tail. Intuitively this makes sense because with the PPI-CPI Gap, both producers and consumers play a similar role, i.e., they are a consumer to someone. On the other hand, if inflation spikes and a business cannot keep up with wages, there can be much more of a divergence. That said, nodes 18, 19, and 17 might be cases for this and further investigation is found in the *Discussion* section.

DISCUSSION

Often with Ball Mapper analyses EDA and visual inspection are the first steps, and then inspection of what points are housed in each node done after. This process of looking into each node helps us understand why we are seeing certain structures and the coloration. To do this investigative work, we combined all of our dimensions in our point cloud (Table 2) with our three derived economic measures (Real Interest Rate, PPI-CPI Gap, Real Wage Growth), as well as the Year, GDP, NBER recession/expansion indication, and a Row ID to join all the data together (Table 5). For the remainder of the discussion we will look closely at each component, starting with C2 and C3. We will spend a significant portion on C1 due to its size and complexity. Then we will turn to the singletons and finish with some questions and future research.

C2

C2 is similarly colored across nodes, but this is misleading because of the average value used for coloration. Nodes 7 and 8 contain the years 1968 and 1969, respectively, and the year 2000 forms the connecting edge. Ball Mapper has paired these years together because of their shared macroeconomic structures. What our analysis suggests is that C2 represents an economic peak that shows a demand-pull inflation structure, reflecting the adage "too much money chasing too few goods."[28] Further, C2 describes a clean or structurally coherent peak, and shows that this kind of peak contains distinct stages.

In Figure 3 through Figure 5, nodes 7 and 8 both show real interest rates approaching 3%, PPI-CPI gaps close to zero, as well as near-zero values for real wage growth. These three metrics indicate the economy was running hot but stable. Prices were not rising disproportionately for either producers or consumers. Both nodes also show strong GDP growth and very tight labor markets where unemployment has been falling through the points in C2. Where C2 splits into stages is that node 7 seems to represent early-stage demand-pull inflation, and node 8 represents late-stage. In node 7 this shows up in the positive housing starts (5.6%), real wage growth (1.3%), moderate inflation (3.7%), and a Fed Rate (6.0%) that has been rising throughout the years. Node 8, however, tells a different story: housing starts roll over (-2.9%), real wage growth shrinking to almost zero (0.17%), inflation continuing to rise (4.3%) and the Federal Reserve aggressively responding with a nominal rate of 7.2%.

The parallel between early and late-stage demand-pull inflation is reflected in distinct historical policy and business choices. The late-1960s demand was primarily supported by mid-1960s fiscally driven policies, spending by the government on the Vietnam war, expansion of the workforce from college-aged graduates, and family formation.[29] In 2000, corporate spending accelerated in anticipation of the digital age, playing a similar demand-driving role to the 1960s policies. This inevitably fueled the "wealth effect", encouraging consumers to drive up demand.[30, 31]

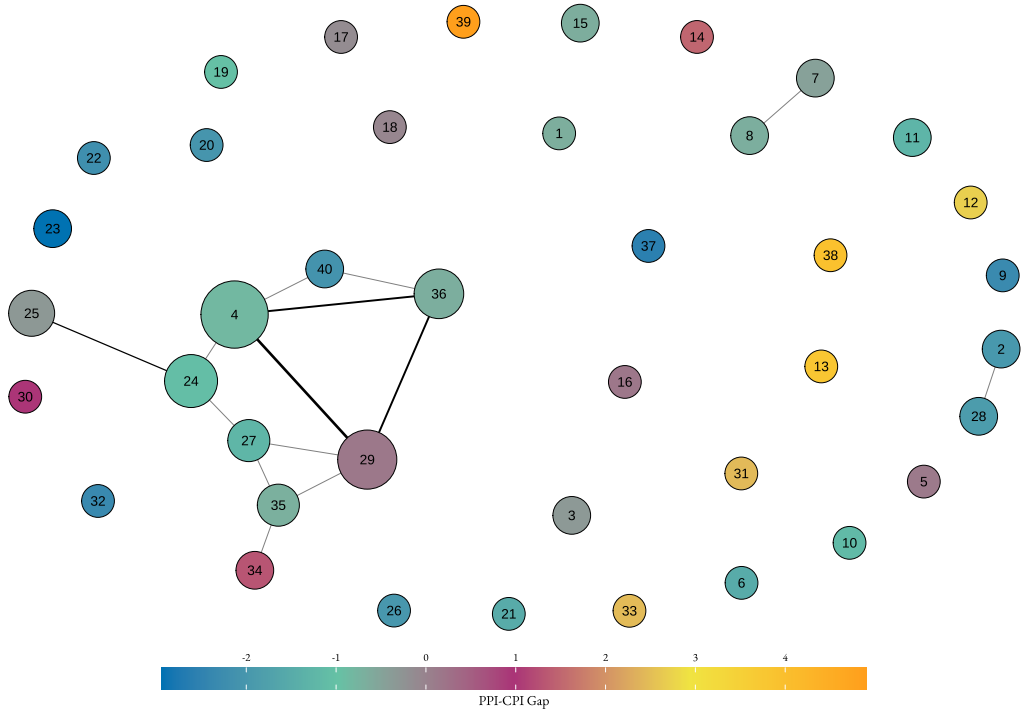


Figure 4: Ball Mapper graph colored by PPI less CPI (PPI-CPI Gap); epsilon set to 0.474.

C3

Similar to C2, node 2 is misleading on the surface in Figure 2, it contains years 1961 and 2002 but the average value places it around the 1980s. Node 28 however is much closer in years, 2001 and 2002. Between the two nodes, though, we notice there is a common theme: they show an economy which is at a trough and has stabilized. What we mean is that this component seems to represent when a contraction has ended, but recovery has not started. Further, unlike C2 we find that C3 shows two different types of troughs: labor market trough and an investment trough. Both nodes show average muted GDP growth (2.2% for node 2; 1.4% for node 28), low Fed Rates (both nodes less than 2.8%) falling throughout the year, housing starts are positive, and inflation is under control (1.3% and 2.2%). However, where they differ is that the labor market seems to break in node 2. Here we see personal income and compensation relatively intact (1.7% and 1.0% respectively), and high unemployment (6.2%) that has been rising being cushioned by elevated transfer utilization. Conversely, node 28 also has elevated unemployment levels, but personal income and compensation are low, causing real wage growth to turn negative (-2.0%). Additionally, domestic investment pulls back aggressively (-3.3%), suggesting business pulling back is the primary driver behind this node.

C1

As mentioned above, what is striking about C1 is that there are two shapes: a triangle and a sweeping arm that runs along the triangle's left side. Although we have not found any literature that explores the relationship between nodes and edge strength (how many points they share), the triangle (denoted body from now on) was something immediately of interest. Beginning with Real Wage Growth, it falls between -2.5% and 2.5% (Figure 5), with an average value of -0.065. A value close to zero tells us that for the points in C1, wages have generally kept up with inflation. The PPI-CPI Gap (Figure 4) shows there is a main group of nodes centered around negative one with three nodes higher (25, 34, and 29), and one lower (40). These mostly negative values suggest that consumer inflation is generally outpacing producers in C1. It's worth noting, however, that our PPI measure does not consider profit margin added, transportation costs included, or other on-the-shelf costs. So this small differential that we see in C1 possibly could be reflecting that gap in pricing, but a more in depth analysis or additional data in our point cloud is needed to provide a more definitive answer. The real interest rate graph, Figure 3, displays a color gradient for C1 that immediately stood out to us. Seeing that it moves from negative to positive values in a semi-orderly manner supports the idea that the nodes with negative values show inflation was outpacing the nominal Fed Rate; and as we move toward the positive values, we see that relationship flip, with the Fed Rate exceeding inflation. Supposing that the Federal Reserve was behind the curve for the negative values and ahead for positive, along with our knowledge that overall real wages have kept up and the PPI-CPI Gap is relatively small, we start to get a picture of what C1 could represent.

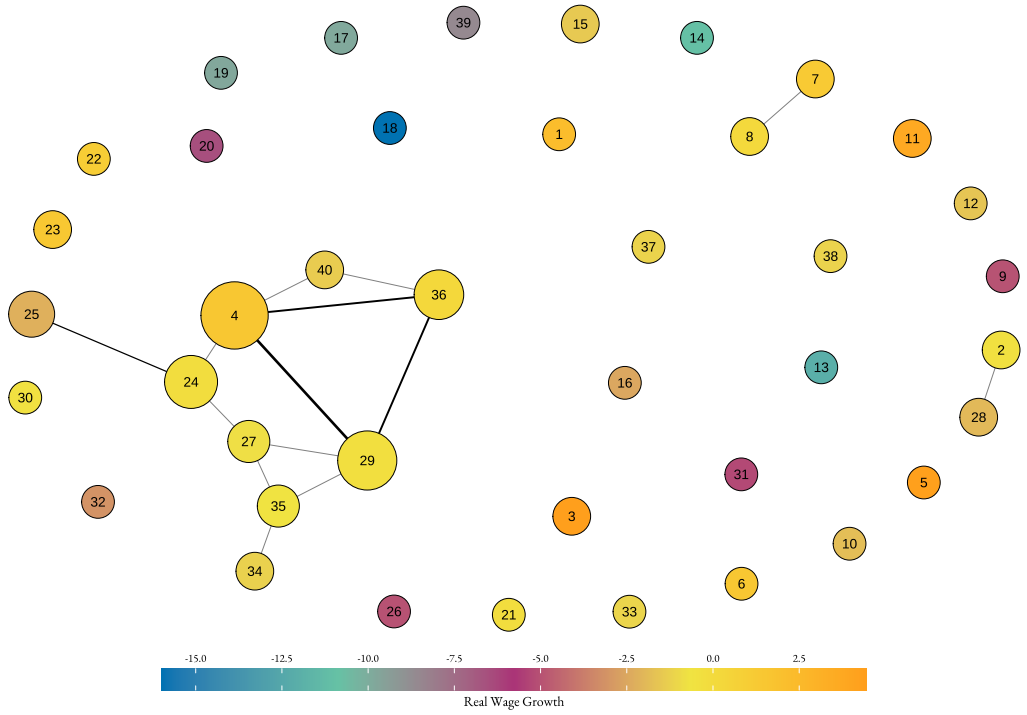


Figure 5: Ball Mapper graph colored by Compensation less Inflation (Real Wage Growth); epsilon set to 0.474.

Indeed, what we have found for C1 is that it represents the U.S. economy at different states of growth. This does not mean that every node in C1 is an instance of good economic conditions, but rather nodes represent certain "phases" that the economy goes through until it turns into a state of overheating expansion (peak) or coming out of a state of contraction (trough). What we have also found is that these peaks and troughs are not included in C1, but are represented by C2 and C3 respectively.

C1: Recovery and Resistance

Broadly, the stages can be categorized in different phases: recovery (34 and 35), stability (29, 4, and 36), resistance (24, 25), and non-canonical (27 and 40). Starting with our recovery phase, we find signs of an economy that has been beaten down but is starting to come back to life. We find that 34 shows a very fragile economy and 35, being connected to more nodes, indicates more stability. The first thing we notice is that housing starts is positive, so builders are feeling confident that they can start projects. The "big four" economic indicators–GDP, Inflation, Unemployment, Fed Rate–on average, show signs of improvement. GDP is increasing, inflation cooling,

Unemployment starting to fall, and the Fed Rate relatively low because they are trying to stimulate the economy. Additionally, real interest rates are negative and real wage growth is improving, suggesting that producers and consumers feel more confident about the economy.

Next, we have what we call the resistance phase. Unlike our recovery phase, nodes 24 and 25 show two different sides of resistance. Node 24 shows an economy which feels like it is still growing and running well. Unemployment, on average, is 5.4% and trending downwards, inflation seems to be under control, around 3.1%. However, at this stage there seems to be some cracks forming in the economy, the first being housing starts turning negative. What is also worrisome is the high Fed Rate (5.9%) and the negative PPI-CPI gap (-1.1%). The combination of these suggests that the Federal Reserve sees that CPI is running ahead of PPI and are concerned about producer inflation catching up to consumer inflation. With CPI outpacing PPI, consumer prices are currently rising, however if producer costs start to rise, the PPI-CPI Gap will start to close and inflation then is a problem for both producers and consumers. To counteract this, the Federal Reserve is preemptively trying to kill inflation for consumers with higher rates. Unfortunately, this has knock-on effects to the labor market restricting wage growth and possibly reversing the direction of unemployment. Node 25 shows the other side of what 24 was warning us about. Housing slows down even more (from close to -5% to -10%, on average), and the PPI-CPI Gap shrinks to almost zero; unlike node 24, producer price inflation is now catching up to consumer price inflation. Since both producers and consumer prices are rising, the Federal Reserve continues to raise rates to an average 7%, sending average real interest rates above 3% and risking the labor market to decline and the economy to possibly enter a recession. We also see real wage growth plummeting from -0.16% (node 24) to -2.23% (node 25). To illustrate this differential, suppose we were given $1,000 at the end of the year and to either had to give some back (negative wage growth), or we received extra (positive wage growth). In node 24 we get that $1,000 and give back $1.60, however in 25 we would have to give back $22.30. So, node 25 shows a "last resort" node before turning into a singleton node. This is similar to 34 in the sense that it is only connected to the other nodes in C1 by one other node, supporting this idea that they are the exit and entrance into economic growth.

Year	PPI Finished Change	Personal Income	Compensation
Entrepreneurship	Transfers	GDP	Consumption
Domestic Investment	Government Spending	Exports	Imports
Inflation	Unemployment	Unemployment Change	Fed Rate
Fed Rate Change	Housing Change	Recession Span (0-1 by 0.25)	Expansion Span (0-1 by 0.25)
Row Number	Real Interest Rate	PPI CPI Gap	Real Wage Growth

Table 5: Data used to review nodes, included additional data that was excluded from out point cloud.

C1: Stability

The final phase tells a story of stability and contains three stages or variations. Starting with node 29, we label it emerging-stability; it shows an economy that has found its legs and is now starting to gain some momentum. Across the three stages, 29 has the highest average unemployment at 5.4% and the lowest average Fed Rate (1.5%). The Federal Reserve (the Fed) is intentionally keeping rates low, making the real interest rate -0.81%, to stimulate the economy. The goal is to make borrowing cheaper to create an incentive to producers and consumers to spend. The PPI-CPI gap at 0.16% informs us that producer prices are slightly growing faster than consumer prices. There are no clear reasons as to why prices are higher on the producer side but some possibilities in the context of emerging-stability are: producers are seeing a pick up in demand or retailers that sell to consumers have not passed on costs to consumers yet in anticipation that conditions get better. Lastly, real wage growth is negative (-0.27%) and consumption is at a moderate 3.08%–the middle average of the three nodes. These two averages support the signs that consumer confidence is gaining and the economy is coming back to life.

The story of node 36 is one of extended-stability. Between the three stages, the producers and consumers seem to be more cautious; on average, the labor market is tightening (unemployment at 4.2%), and GDP (2.5%) along with housing starts (2.4%) are at their lowest. We also see consumer spending slowing down between our three nodes at 2.56%, which is consistent with real wage growth being close to zero (0.24%). The Federal Reserve at this stage has achieved its goal shown by real interest rates at 0.01% and inflation at 2.10%. With its goals met (stable prices and maximum employment), it shifts to actively monitoring for any signs expansion is starting to overheat.

Unlike node 36, node 4 shows the Federal Reserve comfortable with how the economy is running and is letting the economy run its course; we have identified it as full-stability. What gives the Federal Reserve confidence for a "hands off" approach is the low inflation rate (1.94%), unemployment at 5.03%, moderate housing starts (3.96%), and strong GDP growth of 3.72%. Looking at our three derived metrics, our real interest rate is 0.86%, indicating a slight drag on borrowing, allowing the economy to moderate itself as it continues to run efficiently. Our PPI-CPI gap (-0.82%) is telling us consumer prices are rising faster than producer prices. This gap in combination

with the fact that this stage has the highest consumption (3.76%) informs producers that demand is strong at the retail level. Consumers also benefit from real wage growth at 1.56%; their wages are outpacing inflation, giving them an increase in purchasing power. To illustrate using our example for node 25, if we got $1,000 at the end of the year, this positive real wage growth says that we get an extra $15.60.

C1: Non-canonical

Up to this point in our discussion we have seen various stages of the U.S. economy's expansion behavior. Two of our nodes in C1, 40 and 27, do not align with any "phase" we have talked about thus far. Starting with node 40, it is connected to both node 4 and 36; to recall, node 4 is our full-stability and 36 is extended-stability. What we find with node 40 is that consumption is the driving factor. Despite headwinds to the consumer–inflation at 3.48% and real wage growth at -1.38%–they continue to buy. However, housing starts have turned negative (-6.20%), signaling that builders are stepping back and consumers are stepping up. The Federal Reserve, observing consumer resilience and builders' caution, is trying to cool the economy (real interest rates at 1.61%) without crashing the tight labor market (unemployment at 3.83%).

Why we chose to categorize node 27 as non-canonical is because it is the only node which is connected to all three of our phases. Looking at average values, node 27 shows a balancing act of trying to stimulate an economy that is slow to recover without further entrenching sticky, elevated unemployment. On the surface there are signs of confidence. GDP is at 3.33%, there is high domestic investment (8.00%), and housing starts have picked up (10.78%). When looking at the real interest rate, the Fed Rate, and unemployment, though, we see a Federal Reserve trying not to spur inflation while trying to bring unemployment down. On one hand we have a real interest rate of 1.01% that puts a small drag on growth, but on the other the Fed Rate has changed by -1.07% and unemployment is elevated (6.60%) but unchanging (-0.04%).

Singletons

So far we have discussed how our three components represent the various growth stages, structural peaks, and structural troughs for the U.S. economy. As one might expect, singleton nodes do represent states of contraction. However, in addition to the contractions, we found that singletons also represent non-structural, or unstable growth that could feel contractionary. To show that our singletons contain economic contractions we will use a combination of known economic crises as well as observations we found from our derived economic measures. We will also weave in time periods where the economy was growing but in an unusual or unstable way.

Perhaps the most obvious years to show are The Great Recession (2007–2009). Although The Great Recession is known to be '07–'09, nodes 30–32 are the years 2007–2009; however, node 33, another singleton, is the year 2010 seen in Table 4. These years are marked by deep, negative housing starts, very large Fed Rate cuts, and high unemployment. Another more recent time period was the COVID-19 Pandemic. Starting in 2020, the pandemic lasted for a few years, and we see that nodes 37–39 harbor these years: 2020, 2021, and 2022. In 2020, we had a contraction with unemployment rising very quickly and substantial transfers from government stimulus. Following 2020, we had economic expansion in 2021 and 2022 despite people feeling the economy was in a state of contraction. In these two years, we see a pattern of growth that feels bad: positive GDP, falling unemployment, and positive domestic investment mixed with rising inflation, shrinking real wage growth, slowing housing starts, and rapidly rising PPI.

Looking at Table 4, something the reader might notice is that from 1970–1986, nearly every year is a singleton with the exception of 1976–77 and 1985–86. This tells us that Ball Mapper has found meaningful differences between these years such that no year is similar in economic structure to another. Throughout this 16-year period, a few factors that contributed to the distinctiveness are external supply shocks and monetary policy decisions. The oil embargo of 1973–74 was a textbook case of cost-push inflation. This supply shock is visible in Figure 5, where real wage growth reaches one of its lowest values. During 1974 workers' real wages fell to -12.06%. Returning to our previous example, this is receiving that $1,000 but having to give back $120.60 of it.[32]

In 1979 Paul Volcker became the Federal Reserve chairman. Prior to Volcker, the U.S. economy was dealing with the stagflation from the 1970s, characterized by high inflation, rising unemployment, and stagnant growth. So, to eliminate the climbing inflation, Volcker decided to aggressively hike rates with rates peaking in 1981. Specifically, at their peak, Volcker raised the nominal interest rate (Fed Rate) to 16.39% at the expense of the labor market. We see this decision distinctly in Figure 3. Nodes 19–22 represent the years 1981–1984, which show us another example of a contractionary and expansionary period, this time through monetary policy. In 1981, real interest rates were 6.54%, showing that the Federal Reserve was ahead of the curve. By doing this, Volcker achieved his goal of lowering inflation from 9.87% in 1981 to 5.98% in 1982. Raising the Fed Rate, though, made the cost of capital very expensive which caused unemployment to rise and stay around 9.65% from 1982–83. So, during 1981–82 we had a contraction period, and in 1983–84, we saw economic growth with instability in employment and elevated Fed Rates.[33]

Final Notes and Future Research

As the last part of our discussion, there are two primary areas. The first half touches on the limitations we found both with Ball Mapper and the design of our analysis. The second examines some of the unanswered questions that arose during the analysis which are possible avenues for future research.

The primary limitations we had were the choice of time period and number of variables. Initially we used a set of variables we thought would represent the U.S. economy as a whole, but found that they did not all have consistent time ranges. Some data series, such as the Federal Funds Rate, ranged as far back as the early 1900s, and others were only created recently, such as Foreign Direct Investment (2014). While we had 65 years of data, we were not able to capture early structural economic changes or events such as the Great Depression, the New Deal, and WWII. Alternatively, we could have chosen a different frequency of data, such as monthly or quarterly. This would have given us more points in our point cloud and perhaps a more nuanced picture of the U.S. economy. Our analysis also could have included more dimensions, testing Ball Mapper's ability to handle a multitude of dimensions. It also could have asked a more focused economic question to see if there is a nuanced structure in a particular area of the economy.

Focusing on the technical limitations, the first is that the standard Euclidean Distance between points is the only distance metric used. Currently, in Ball Mapper (R version) there is no way to easily change the distance metric. To do so one would need to copy the source code of the *BallMapper* function and either: adapt the code to add an argument for a distance metric, or hard code it into a copy of the original function. Additionally, the *BallMapper* function in R uses a greedy, sequential algorithm to select ball centers. When a landmark is chosen, the Ball Mapper algorithm does so in a sequential top-down order. This rigid structure is good when others would like to verify the output of the graph; having the same dataset, in the same order, using the same epsilon, will produce the exact same graph. However, Ball Mapper could be susceptible to seasonal data where you have large naturally occurring spaces between the data. While seasonality is a valid characteristic of a dataset, it might not be the true underlying structure. Because of Ball Mapper's greedy, sequential algorithm, it may pick up on the seasonality of the data right away rather than comprehensively looking at the dataset. To fix some of these issues, one could randomize the dataset every time Ball Mapper is run, or make sure to seasonally adjust or normalize any data. Currently, some of these fixes have been implemented in the package *pyBallMapper*, written in Python. In this package they have added an order argument to allow randomization of rows. They also have included a distance metric argument so a user can choose what distance metric they would like to use. Finally, *pyBallMapper* uses Python's NumPy package which utilizes vectorization, speeding up the sequential nature of the Ball Mapper algorithm.

To start the second half, we noticed during the analysis there is not a clear way to know whether Ball Mapper is giving us a stable graph of our data. We discussed that if given the same data, in the same order, it indeed will give us the same graph. However, we did not find any literature answering this exact question. The closest paper we could find was one which studied the Brexit Vote using Ball Mapper. At the end of this paper they wrote about a robustness test they implemented by showing why the vote to leave the European Union passed.[34] What is unclear is how they ran a stated 10,000 repetitions. From what we know about the Ball Mapper algorithm, we could infer that they randomize the order of their data every time, or took infinitesimal slices of an interval such that theoretically Ball Mapper would produce a different graph. The former would be a good standard to show general robustness for any Ball Mapper output and could look like running many repetitions while randomizing the order of the data every time, and simultaneously collecting summary statistics of each graph. Some example statistics might include the number of components, nodes, and edges at a singular epsilon or a range of epsilons. One could also look at edges and edge weights for consistency, as well as some sort of base value, noting if there is a large or small variance.

Moving to more abstract questions that came up during our analysis, our questions revolved around nodes' relationships to each other based on various factors: edge strength, connectedness (degree), and node size. Acknowledging that node relationships could have a combination of these factors, our primary questions are: does edge strength between nodes indicate a stronger relationship? For our graph, does the edge strength between nodes 4 and 29 represent a stronger structural signal than the edge between node 25 and 24? There is also the question of connectedness or degree–how many edges a node has. Similar to edge strength, does the fact that node 27 has three edges indicate its importance over node 34 with just one edge? Lastly, and in combination with the previous two, does node size carry any relational value? Using our node 4 and node 29 example again, since they are the largest nodes with the strongest edge weight, does this combination indicate that their relationship is stronger? That is, if we were to look at this graph over many epsilons, would the connection between nodes 4 and 29 persist while the connection between nodes 4 and 40 does not? All of these questions came up during our analysis, and to our knowledge have not been studied; they are all areas for future research on Ball Mapper.

CONCLUSIONS

Although there is much further work to do on developing Ball Mapper, through our analysis we have shown how TDA, and Ball Mapper in particular, is able to give insight into data that we might not see or describe easily. Throughout this paper we have hopefully guided our reader to better understand how Ball Mapper works, as well as shown a path to start implementing it. From our structural map of the U.S. economy from 1960 to 2024, Ball Mapper revealed structural expansion, structural "clean" peaks and troughs, as well as showing contractions and non-organic growth. By displaying the Ball Mapper output with a new graph, we were able to clearly explain how to systematically take high-level, global interpretations of our Ball Mapper output and use them in combination with lower-level, local, node-specific investigation. We also explained some of the limitations we discovered and touched on future work based on questions that came up for us during the analysis.

Ball Mapper has been applied to many fields ranging across the social sciences, finance, and the life sciences. Ball Mapper, and TDA in general, may have an advantage in some of the social sciences because it can combine both qualitative and quantitative data in the same analysis. For instance, in our Brexit example from our *Discussion*, the authors sought to understand qualitative and quantitative motivations behind the Brexit vote outcome. Some of the variables they used were Household composition (single, married, etc.), Car ownership (number of cars), Housing Tenure (Owned, Private rental, etc.), and Age. Whether Ball Mapper brings new insights, or shows what is known in a unique way, we hope that this paper serves as an accessible entry point for researchers of all disciplines who are interested in what TDA can offer.

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Ryan Z. Johnson is an alumnus of the University of Alaska - Anchorage. They graduated spring of 2018 with a Bachelor's of Science in Mathematics. During their senior year, Ryan was urged by their advisor Dr. Mark Fitch to pursue TDA after doing presentation for a course and connected with Dr. Samuel Cook. Since then, they have continually worked with Dr. Cook outside their profession in social services. They will be continuing their education at the University at Albany as a graduate student in the Master's in Data Science program. This paper provided valuable preparation and experience for the upcoming program's concentration areas: TDA, machine learning, and statistics.